

Meta-analysis of the effects of ractopamine hydrochloride on carcass cutability and primal yields of finishing pigs.



The objective was to summarize previous literature, using a meta-analysis approach, on the effects of ractopamine hydrochloride (RAC) when fed at doses of 5 to 10 mg/kg for up to 35 d before harvest on carcass cutability and belly quality of finishing pigs. The meta-analysis provided an opportunity to determine the consensus of previously published literature. Ten studies were evaluated to determine cutting yields and 8 studies were used to determine belly quality in this review. Pooled dietary RAC concentrations (5 mg/kg, 7.4 mg/kg, 10 mg/kg, and step-up feeding programs) and pooled feeding durations (up to 35 d before harvest) were compared with pigs not fed RAC (controls) and were analyzed as a meta-analysis using the mixed procedure of SAS. Ractopamine inclusion was the fixed effect in the model and the individual study was considered a random variable. The only difference between RAC and control pigs for whole primals as a percentage of side weight was the whole ham ($P < 0.01$). No other differences were detected for whole primals as a percentage of side weight. Yet, differences were detected in the standardized trimmed primal yields. A difference ($P < 0.05$) in percentages of the side weight was detected for the Boston butt, trimmed loin, and trimmed ham. This translated into RAC pigs having a carcass cutting yield (74.70% vs. 73.69%, respectively; $P = 0.02$; SED = 0.33) advantage of 1.01% units and a bone in lean cutting yield (61.43% vs. 60.33%, respectively; $P = 0.03$; SED = 0.40) advantage of 1.10% units when compared with control pigs. The advantage in bone-in cutability was a result of increased boneless sub primal yields in each of the lean cuts (shoulder, loin, and ham). When further evaluated, RAC pigs had a boneless shoulder (Boston butt + picnic) yield advantage of 0.32% units ($P < 0.01$; SED = 0.11), a 0.43% unit ($P = 0.01$; SED = 0.13) yield advantage in the boneless loin (Canadian back + tenderloin + sirloin), and a 0.51% unit ($P < 0.001$; SED = 0.11) advantage in the boneless ham (inside + outside + knuckle). A boneless yield was calculated using a summation of the percentage of side weight from the boneless shoulder, boneless loin, and boneless ham, which resulted in a 1.08% unit (36.28% vs. 35.20%, respectively; $P = 0.002$; SED = 0.25) advantage of RAC pigs when compared with control pigs. There were no subprimal yield differences ($P = 0.93$) in the trimmed belly between RAC pigs (12.18%) and control pigs (12.18%). However, RAC pigs (15.27 cm; 73.42) had narrower flop distances ($P = 0.02$; SED = 0.62) and greater iodine values ($P = 0.01$; SED = 0.33), respectively, when compared with control pigs (17.08 cm; 71.48).